CF HW 4

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Exercise 3

We obtain the BS closed-form option price 0.377. We used formula (3.23) with constant volatility $\sigma_* = 0.57$, as derived in (2.25), and an average interest-rate 0.065.

We implement Monte Carlo simulation with Milstein and Euler discretization as described in Chapter 9. Figure 1 and 2 show the convergence of the simulated prices to the analytical price as the number of simulated paths increases. The only difference is that the Milstein and Euler paths in Figure 2 are simulated using the same Brownian motion.

It is not particularly clear from Figure 1 and 2 that the convergence improves as the number of time steps increases. Both Euler and Milstein discretizations seem to have similar performance from our numerical analysis. However, Milstein scheme improves the speed of convergence compared to the Euler discretization according to the book.

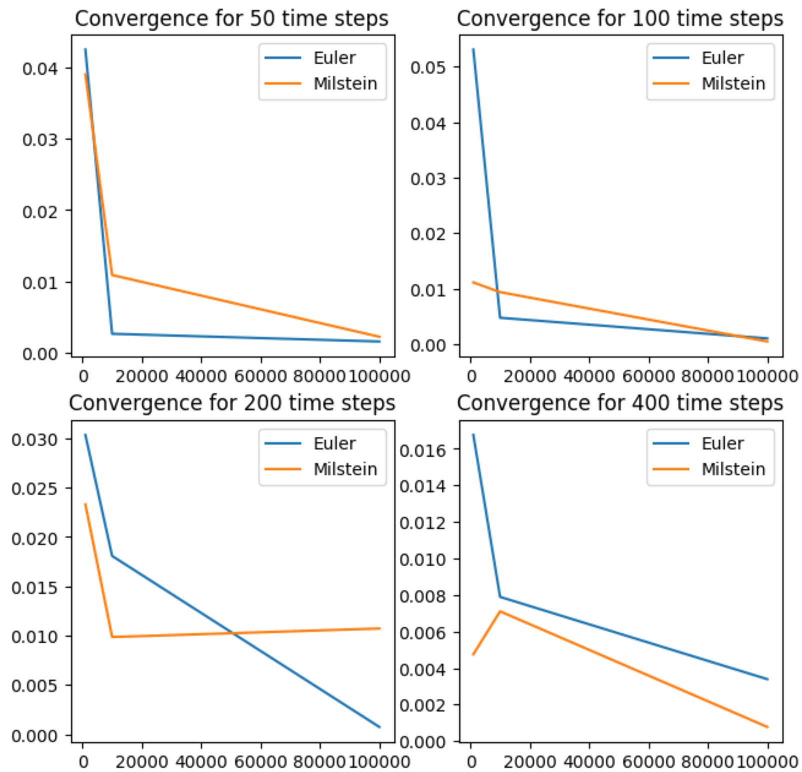

Figure 1: Convergence of the absolute difference between simulated and analytical prices to zero using different Brownian motions in the Euler and Milstein paths.

Up-and-Out Barrier Option becomes worthless if the price of the underlying asset rises above a certain barrier level B at any time during its life. If the barrier is not breached, the option behaves like a standard European option.

The simulated up-and-out barrier option price is 0. This makes sense because we only receive a payoff if the asset value is greater than $K = 1.6$, but the option is worthless in that case since the barrier value $B = 1.5$ has been reached. Hence, we are completely certain about the zero price.

Up-and-In Barrier Option only becomes active if the price of the underlying asset rises above a certain barrier level B at any time during its life. If the barrier is breached, the option behaves like a standard European option from that point onward.

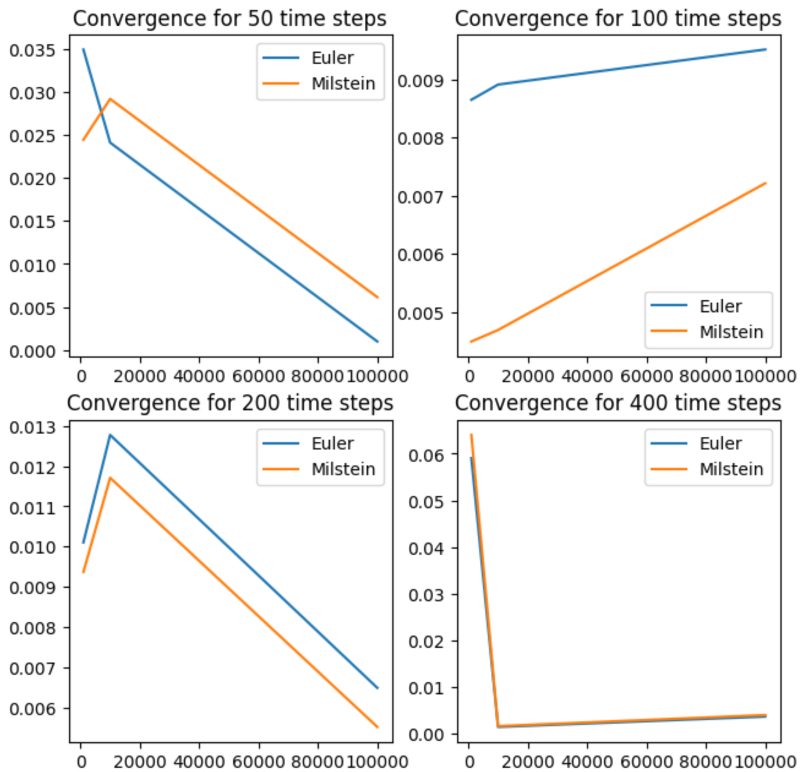

Figure 2: Convergence of the absolute difference between simulated and analytical prices to zero using the same Brownian motion in the Euler and Milstein paths.

The simulated up-and-in barrier option price is 0.374. Since $B < K$, the pricing of this option is identical to the pricing of the European option. The barrier has no effect on our payoff because an inactive option has not reached value K, and hence, gives zero payoff anyway. Therefore, we know that the closed-form option price is 0.377. That means that the simulated option price is very accurate (error = 0.003).

Exercise 4

When $\rho = -0.9$, we obtain an estimate 1.41 and a 0.95 CI [1.4044, 1.4139]. Similarly, when $\rho = 0.9$, we obtain an estimate 1.21 and a 0.95 CI [1.2082, 1.2185]. We observe that the option price for negatively-correlated Brownian motion is greater than the price for positively-correlated Brownian motion.

When ρ is negative, $S_1(t)$ and $S_2(t)$ tend to move in opposite directions. This means that if $S_1(T)$ is high, $S_2(T)$ is likely to be low, and vice versa. Therefore, the maximum of $S_1(T)$ and $S_2(T)$ will be more variable, often capturing higher values from the individual asset prices.

Mathematically, for $\rho < 0$, the distribution of $\max(S_1(T), S_2(T))$ is more spread out, increasing the tail probabilities of higher payoffs. Thus, the expected payoff $E[\max(S_1(T), S_2(T))]$ is higher.

To see this more rigorously, consider the joint log-normal distribution of $S_1(T)$ and $S_2(T)$. The expected payoff $E[\max(S_1(T), S_2(T))]$ can be approximated or numerically integrated given the joint distribution. For positive correlation, the joint distribution is tighter, leading to lower expected maximum values. For negative correlation, the joint distribution is wider, leading to higher expected maximum values.

Exercise 5

a.

Figure 3 shows the Euler discretization of the paths of $v(t)$. Notice that the red path reaches value -0.003 at time 2.2.

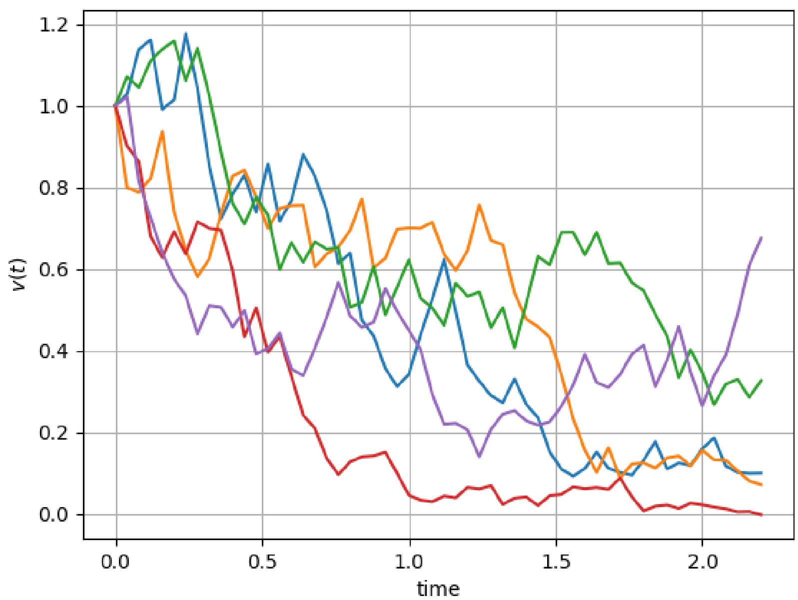

Figure 3: Euler discretization of the paths of $v(t)$.

b.

Figure 4 and 5 show that the AES paths are nonnegative when sampling from the non-central chi-squared distribution. Figures 7 and 8 show the expectation and variance.

c.

Figure 6 shows Euler discretziation of the paths of $v(t)$ with truncation. Figures 7 and 8 compare the expectation and variance of the truncated Euler paths and AES paths. We could implement the exact variance and expectation and then compare which path simulation better matches the theoretical expectation and variance.

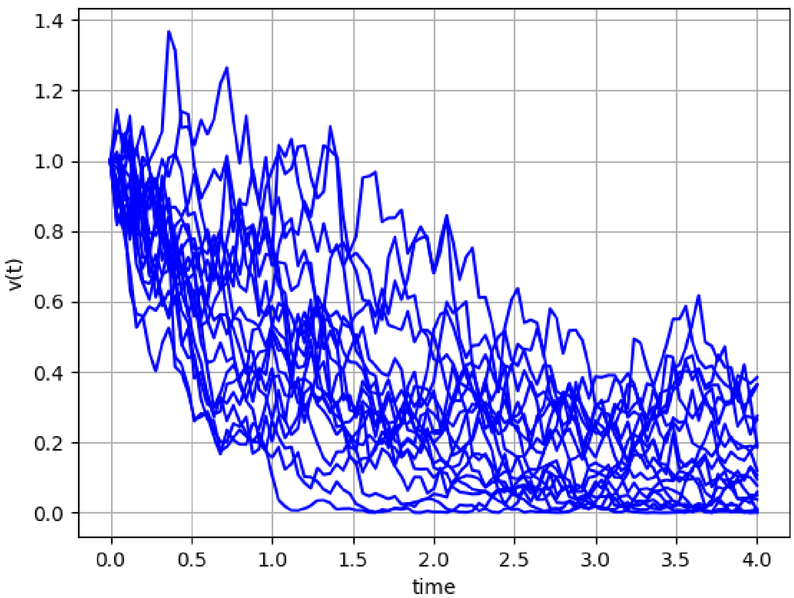

Figure 4: AES paths of $v(t)$ for 100 time steps.

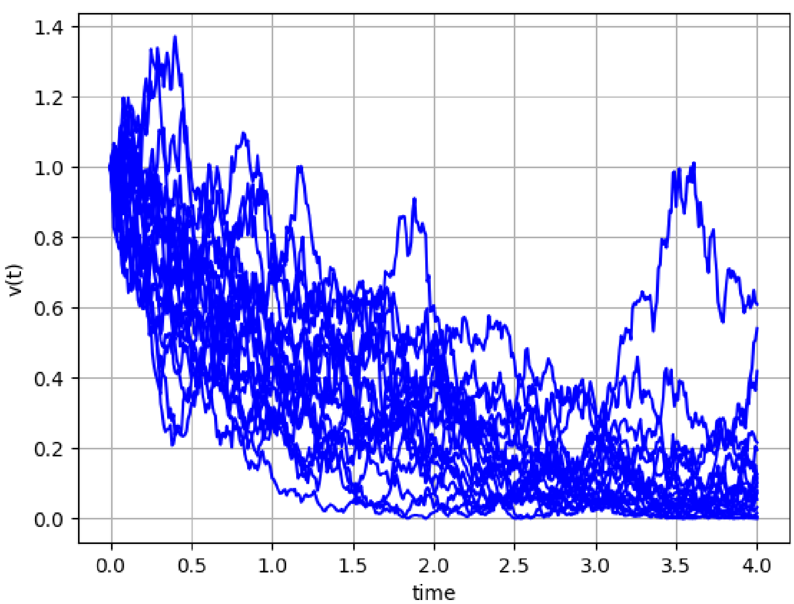

Figure 5: AES paths of $v(t)$ for 400 time steps.

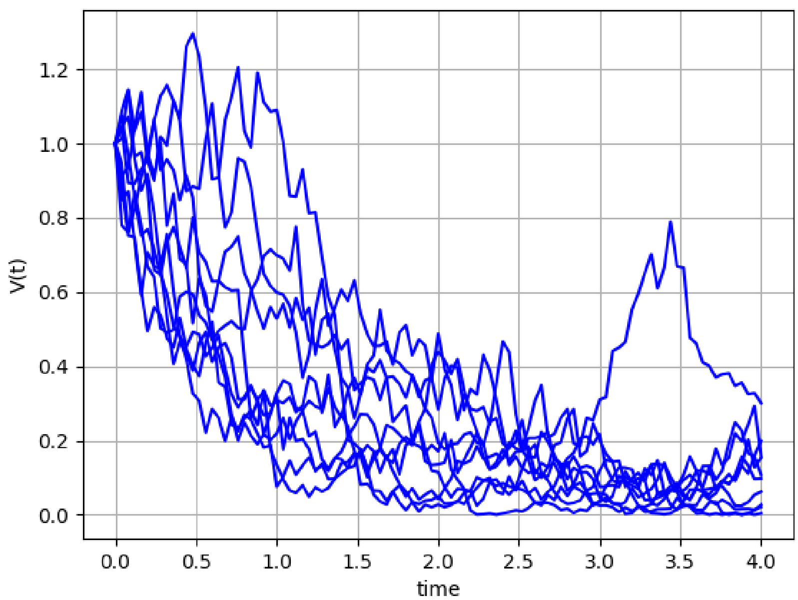

Figure 6: Euler paths of $v(t)$ with truncation.

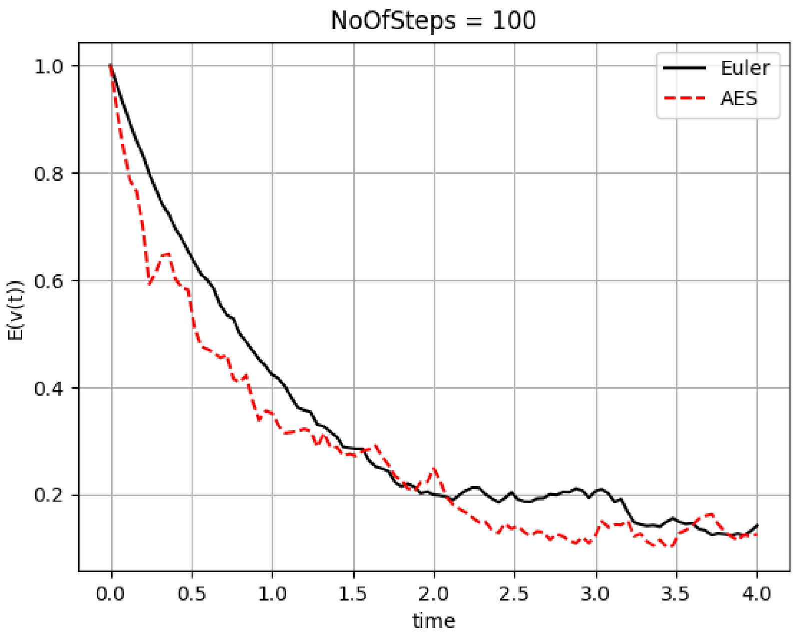

Figure 7: Mean comparison.

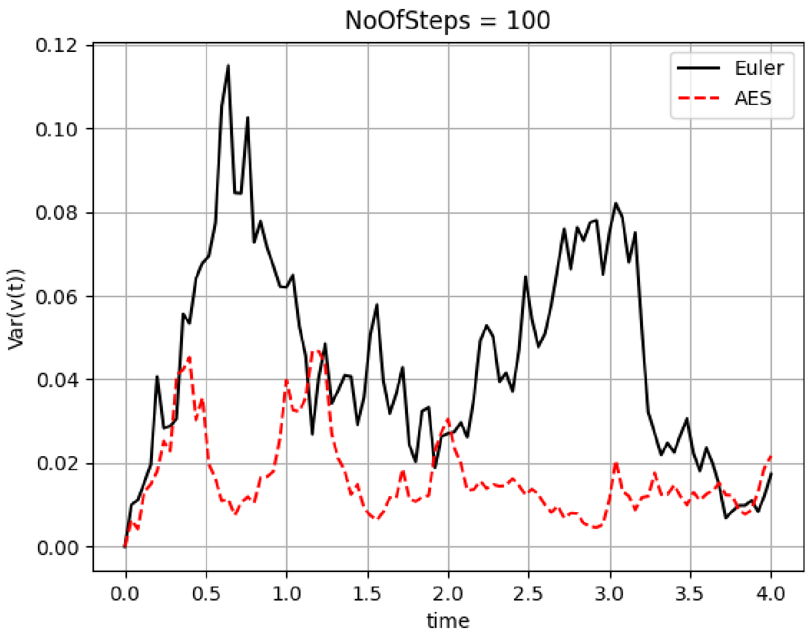

Figure 8: Variance comparison.